# MCS4302 Week 6 Lab - Measurement Sheet

Index number: 24440191 | Repository: https://github.com/Deshan-Ruhunage/mcs4302-running-system |

Device: Android emulator (Pixel 7, API 37)

## Part 1 - flutter test passing

```
deshanruhunage@Deshans-MacBook-Air mobile_client % cd ~/Documents/MCS4302/week4/mobile_client
flutter test
Resolving dependencies...
Downloading packages...
  cupertino_icons 1.0.9 (2.0.0 available)
  flutter_local_notifications 22.3.2 (22.3.3 available)
  material_color_utilities 0.13.0 (0.13.1 available)
  meta 1.18.3 (1.19.0 available)
  petitparser 7.0.2 (7.1.0 available)
  test_api 0.7.12 (0.7.15 available)
  vector_math 2.4.0 (2.4.4 available)
  vm_service 15.3.0 (15.4.0 available)
  web 1.1.1 (1.1.2 available)
  xml 7.0.1 (7.1.0 available)
Got dependencies!
10 packages have newer versions incompatible with dependency constraints.
Try `flutter pub outdated` for more information.
00:05 +7: All tests passed!
deshanruhunage@Deshans-MacBook-Air mobile_client %
```

TODO 1 (404 not retried) and TODO 2 (error state) pass. Check: adding || code == 404 to isRetryableStatus made the 404 test fail (+3 −1); reverted.

## Part 2 – Seeded bug

**Error:** setState() called after dispose(): _ItemDetailPageState (lifecycle state: defunct, not mounted)

**File and line:** lib/week6/detail_page.dart:34 – the setState call in _ItemDetailPageState._load()

**Cause:** the item request completed after the user pressed Back, so _load() called setState() on a State that was already disposed (mounted = false). Fixed with if (!mounted) return; before setState; the regression test now passes.

## Part 3 – Performance (profile mode, mean ± SD of 3 runs)

**Prediction:** Isolate.run will drop fewer frames, but will not give a shorter end-to-end time, because it adds worker start-up and copying.

| n | Decode on | End-to-end (ms) | Decode (ms) | Worst frame (ms) | Est. dropped frames |
|---|---|---|---|---|---|
| 5k | UI isolate | 556 ± 232 | 101 ± 54 | 190 ± 54 | 19 ± 6 |
| 5k | Isolate.run | 377 ± 59 | 84 ± 38 | 177 ± 40 | 21 ± 13 |
| 50k | UI isolate | 3106 ± 286 | 353 ± 60 | 549 ± 66 | 48 ± 7 |
| 50k | Isolate.run | 3235 ± 336 | 711 ± 108 | 178 ± 10 | 51 ± 7 |

**Spinning icon:** at 50k the icon stuttered a little with UI isolate, and kept spinning with Isolate.run. All runs used the decode window.

## Interpretation

Isolate.run improved the worst frame at 50k (549 → 178 ms), so the UI no longer paused, matching the icon. It did not improve end-to-end time (3106 vs 3235 ms), which is dominated by the ~2.7 s fetch, and decode time roughly doubled (353 → 711 ms) because it includes starting the worker and copying the 9.5 MB string; the dropped-frame estimate did not fall (48 vs 51) because the longer decode window counts more of the emulator's background jank. At n = 5k the decode is only ~100 ms and the differences are within the run-to-run variation, so Isolate.run was not

worth it there. My prediction was right about end-to-end time but only partly right about frames: the worst frame improved, the dropped-frame estimate did not. Emulator numbers compare conditions, not real-phone values.

## Appendix – W6RUN console lines

Format: W6RUN, mode, n, end-to-end ms, fetch ms, decode ms, worst frame ms, est. dropped frames, window. The two warm-up runs (not recorded) were: W6RUN, ui,5000,556,518,37,74.0,11,4, decode and W6RUN, ui,50000,4206,3904,301,728.0,51,3, decode.

```
W6RUN,ui,5000,455,333,121,249.5,22,3,decode
W6RUN,ui,5000,392,351,40,142.7,12,2,decode
W6RUN,ui,5000,821,678,143,179.1,23,3,decode
W6RUN,isolate,5000,309,234,75,157.3,14,2,decode
W6RUN,isolate,5000,417,292,125,222.3,36,4,decode
W6RUN,isolate,5000,404,352,51,150.5,13,2,decode
W6RUN,ui,50000,3275,2931,343,575.4,48,4,decode
W6RUN,ui,50000,2776,2358,417,598.0,55,4,decode
W6RUN,ui,50000,3266,2967,299,474.2,42,4,decode
W6RUN,isolate,50000,3239,2620,619,178.3,48,7,decode
W6RUN,isolate,50000,2897,2067,830,188.4,59,8,decode
W6RUN,isolate,50000,3568,2885,683,167.9,47,7,decode
```